

Extended Euclidean Algorithm (EEA) Methods

We used the Extended Euclidean Algorithm (EEA) to find that **271** is the inverse of 71 mod 481 using the standard (easy to remember) Q, R, s, t method.

Each row contains a linear combination of 418 and 71 that is equal to the remainder: $418s + 71t = R$

EEA standard:

	Quotient	Remainder	s	t
$481 \bmod 71 = 55$	6	55	1	-6
$71 \bmod 55 = 16$	1	16	-1	$1 - 1(-6) = 7$
$55 \bmod 16 = 7$	3	7	$1 - 3(-1) = 4$	$-6 - 3(7) = -27$
$16 \bmod 7 = 2$	2	2	$-1 - 2(4) = -9$	$7 - 2(-27) = 61$
$7 \bmod 2 = 1$	3	1	$4 - 3(-9) = 31$	$-27 - 3(61) = -210$

$$31(481) - 210(71) = 1$$

$$-210 \equiv \mathbf{271} \bmod 481$$

Adding two initial rows for 481 and 71, and switching to non-modular notation allows for a fully recursive fast method for computing the remaining rows:

EEA_fast:

	s	t
481	1	0
71	0	1
$(481) - 6(71) = 55$	$1 - 6(0) = 1$	$0 - 6(1) = -6$
$(71) - 1(55) = 16$	$0 - 1(1) = -1$	$1 - 1(-6) = 7$
$(55) - 3(16) = 7$	$1 - 3(-1) = 4$	$-6 - 3(7) = -27$
$(16) - 2(7) = 2$	$-1 - 2(4) = -9$	$7 - 2(-27) = 61$
$(7) - 3(2) = 1$	$4 - 3(-9) = 31$	$-27 - 3(61) = -210$

$$31(481) - 210(71) = 1$$

$$-210 \equiv 271 \pmod{481}$$

The EEA_fast method can also be easily implemented in Python:

```
In [4]: #Extended Euclidian Algorithm
#Returns s,t where sa + tb = GCD(a,b)

def EEA_fast(a,b):
    s=[1,0]
    t=[0,1]
    n=1
    while (a%b)!=0:
        n+=1
        q=a//b
        r=a%b
        s_n=s[n-2]-q*s[n-1]
        t_n=t[n-2]-q*t[n-1]
        s.append(s_n)
        t.append(t_n)
        print(s_n,t_n)
        a=b
        b=r
    return (s[n],t[n])
```

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In [5]: EEA_fast(481,71)
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```
1 -6
-1 7
4 -27
-9 61
31 -210
```

```
Out[5]: (31, -210)
```

The classic (i.e. textbook) method involves letting $a=481$ and $b=71$, and writing each row as a linear combination of a and b that is equal to the remainder:

EEA classic:

$$481$$

$$a=481$$

$$71$$

$$b=71$$

$$(481) - 6(71) = 55$$

$$(a) - 6(b) = 55$$

$$a-6b=55$$

$$(71) - 1(55) = 16$$

$$(b) - 1(a-6b) = 16$$

$$-a+7b=16$$

$$(55) - 3(16) = 7$$

$$(a-6b) - 3(-a+7b) = 7$$

$$a-6b+3a-21b=7$$

$$4a-27b=7$$

$$(16) - 2(7) = 2$$

$$(-a+7b) - 2(4a-27b) = 2$$

$$-a+7b-8a+54b=2$$

$$-9a+61b=2$$

$$(7) - 3(2) = 1$$

$$(4a-27b) - 3(-9a+61b) = 1$$

$$4a-27b+27a-183b=1$$

$$31a-210b=1$$

$$31(481)-210(71)=1$$

$$-210 \equiv 271 \pmod{481}$$